

Networking Fundamentals
BY
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1. Introduction

Computer networking is the process of connecting two or more computers or devices so that they can communicate and share information. The connection may be wired, such as through Ethernet cables, or wireless, such as through Wi-Fi. A network allows devices to share files, use common services, access the internet, and communicate with one another.

Networking knowledge is very important in cybersecurity because most attacks and security incidents happen through a network or involve network communication. A cybersecurity professional should understand how data travels, how devices are identified, how services communicate, and how attackers may scan or misuse network resources. Understanding networking also helps in identifying suspicious traffic, configuring firewalls, monitoring systems, and investigating security incidents.

The TryHackMe learning activities related to this report introduce basic networking concepts and the fundamentals of local area networks. The topics help beginners understand common networking terms, how devices communicate, and how a LAN is arranged using devices such as switches and computers. These concepts are useful as a foundation for later cybersecurity topics such as network scanning, packet analysis, and network defence.

Objectives of the Report

- Understand the meaning of computer networking.
- Learn basic networking terms and communication methods.
- Understand the purpose and components of a LAN.
- Connect networking concepts with practical cybersecurity learning.

This report is written in simple language as a basic student-level explanation of the networking topics.

2. What is Networking?

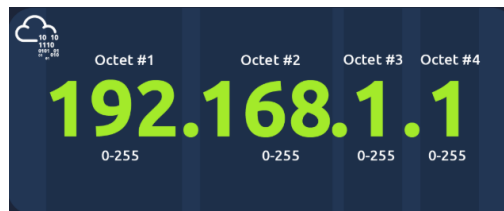
A network is a group of connected devices that can exchange data and share resources. The devices can include computers, mobile phones, printers, servers, routers, and smart devices. Networks may be small, like the network in a home, or very large, like the internet. The first iteration of the Internet was within the ARPANET project in the late 1960s. This project was funded by the United States Defence Department and was the first documented network in action.

Client: A client is a device or application that requests a service. For example, a browser on a laptop is a client when it requests a website.

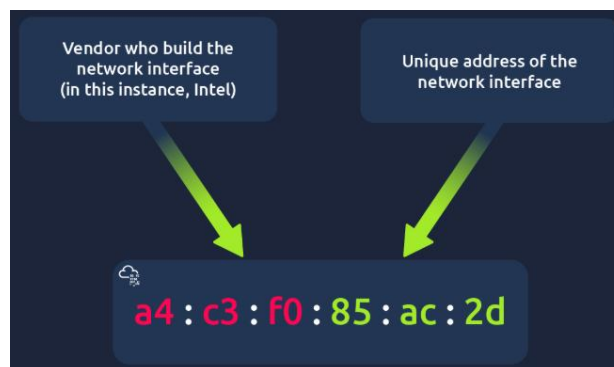
Server: A server is a computer or system that provides a service or resource to clients. A web server provides web pages to users.

Packet: A packet is a small unit of data sent across a network. Large information is divided into smaller pieces so that it can be transmitted efficiently.

IP Address: An IP address is a logical address used to identify a device on an IP network. IPv4 addresses are commonly written in a format such as 192.168.1.10. The value of each octet will summarise to be the IP address of the device on the network. IP Addresses follow a set of standards known as protocols. These protocols are the backbone of networking and force many devices to communicate in the same language. Depending on where they are will determine what type of IP address they have: a public or private IP address. A public address is used to identify the device on the Internet, whereas a private address is used to identify a device amongst other devices.



MAC Address: A MAC address is a hardware-related address associated with a network interface. It is generally written in hexadecimal format split into two's and separated by a colon, such as 00:1A:2B:3C:4D:5E. However, an interesting thing with MAC addresses is that they can be faked or "spoofed" in a process known as spoofing. This spoofing occurs when a networked device pretends to identify as another using its MAC address.



Ping: Ping is a network utility used to check whether a device is reachable. It normally sends an ICMP (Internet Control Message Protocol) echo request and waits for a reply. If a user cannot access another computer, ping can help check whether the destination is reachable. A successful reply suggests that the device responded, while a failed reply may indicate a connectivity problem, firewall restriction, incorrect address, or unavailable device.

IP Address vs MAC Address

IP Address	MAC Address
Logical address used at the network layer	Hardware/interface address used at the data-link layer
Can change depending on the network	Usually assigned to the network interface
Helps route data between networks	Helps identify interfaces on a local network
Example: 192.168.1.10	Example: 00:1A:2B:3C:4D:5E

How Two Computers Communicate

When two computers communicate, the sender first prepares the information that needs to be transmitted. The information is divided into packets. The sender uses addressing information so that the packets can reach the correct destination. Network devices such as switches and routers help forward the traffic.

Real-World Example: Opening a Website

When a student opens a website, the laptop acts as a client. It sends a request through the local Wi-Fi router and the internet. The request reaches a web server, which sends back the requested web page. The page is transferred as network data and displayed in the browser. Although the process appears immediate, many networking steps happen in the background.

The lab covers various tasks which are described below with what the tasks are and what we are supposed to do in these.

TASK 1: What is Networking?

Answer the questions below

What is the key term for devices that are connected together?

✓ Correct Answer

TASK 2: What is Internet?

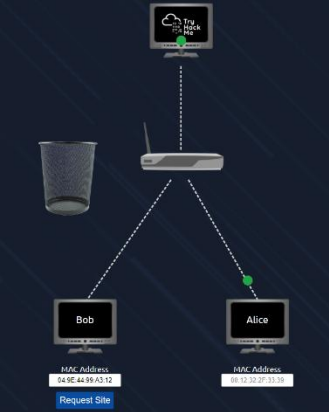
Answer the questions below

Who invented the World Wide Web?

✓ Correct Answer 

TASK 3: Identifying Devices on a Network

The lab task that is assigned in the room is to access hotel WI-FI network which is paid. The router is not allowing access if you send request for connection but the other person is able to access, so I tried changing the MAC for the device with the other one.



What does the term "IP" stand for?

✓ Correct Answer

What is each section of an IP address called?

✓ Correct Answer

How many sections (in digits) does an IPv4 address have?

✓ Correct Answer 

What does the term "MAC" stand for?

✓ Correct Answer

Deploy the interactive lab using the "View Site" button and spoof your MAC address to access the site. What is the flag?

✓ Correct Answer

TASK 4: PING (ICMP)

The lab task for ping to check the network connectivity with the other IP address with the command prompt using ping command.

What protocol does ping use?

✓ Correct Answer

What is the syntax to ping 10.10.10.10?

✓ Correct Answer

What flag do you get when you ping 8.8.8.8?

✓ Correct Answer

3. Introduction to LAN

LAN stands for Local Area Network. It is a network that connects devices within a limited geographical area, such as a home, office, school, laboratory, or building. A LAN usually provides fast communication and allows users to share resources such as printers, files, applications, and internet access.

Purpose of a Switch

A switch is a networking device used to connect multiple devices in a LAN. It receives data frames and forwards them toward the intended destination. A switch learns the MAC addresses of connected devices and uses this information to send traffic through the appropriate port instead of sending every frame to every device.

Common Components of a LAN

- Computers and laptops: End-user devices that send and receive data.
- Switch: Connects devices inside the local network.
- Router: Connects the LAN to other networks, including the internet.
- Access point: Provides wireless connectivity to devices.
- Ethernet cables: Carry data between wired devices.
- Network interface card: Allows a device to connect to a network.
- Printer or shared storage: Resources that can be used by multiple users.

Network Topology

Network topology means the physical or logical arrangement of devices and connections in a network. One common topology is the star topology. In a star topology, devices are connected to a central device, usually a switch. If one cable connected to an individual computer fails, the other computers can usually continue working. However, failure of the central switch can affect the whole LAN.

Subnetting

Subnetting is the term given to splitting up a network into smaller, miniature networks within itself. Think of it as slicing up a cake for your friends. Network administrators use subnetting to categorise and assign specific parts of a network to reflect this. Subnetting is achieved by splitting up the number of hosts that can fit within the network, represented by a number called a subnet mask.

Subnets use IP addresses in three different ways:

- Identify the network address
- Identify the host address
- Identify the default gateway

Type	Purpose	Explanation	Example
Network Address	This address identifies the start of the actual network and is used to identify a network's existence.	For example, a device with the IP address of 192.168.1.100 will be on the network identified by 192.168.1.0	192.168.1.0
Host Address	An IP address here is used to identify a device on the subnet	For example, a device will have the network address of 192.168.1.1	192.168.1.100
Default Gateway	The default gateway address is a special address assigned to a device on the network that is capable of sending information to another network	Any data that needs to go to a device that isn't on the same network (i.e. isn't on 192.168.1.0) will be sent to this device. These devices can use any host address but usually use either the first or last host address in a network (.1 or .254)	192.168.1.254

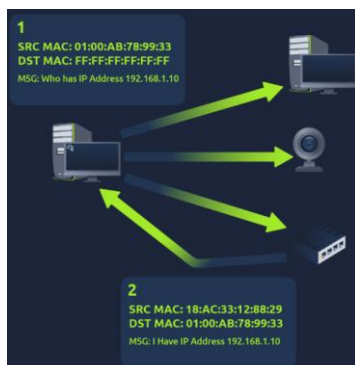
ARP

Address Resolution Protocol is the technology that is responsible for allowing devices to identify themselves on a network.

Each device within a network has a ledger to store information on, which is called a cache. In the context of ARP, this cache stores the identifiers of other devices on the network.

In order to map these two identifiers together (IP address and MAC address), ARP sends two types of messages:

1. ARP Request
2. ARP Reply



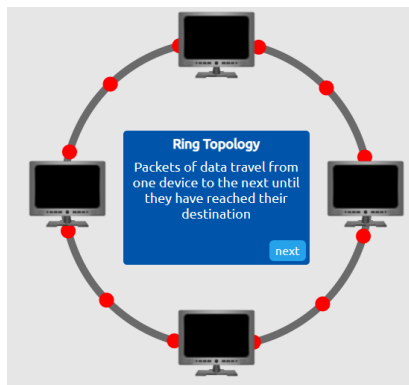
DHCP

IP addresses can be assigned either manually, by entering them physically into a device, or automatically and most commonly by using a DHCP (Dynamic Host Configuration Protocol) server. When a device connects to a network, if it has not already been manually assigned an IP address, it sends out a request (DHCP Discover) to see if any DHCP servers are on the network. The DHCP server then replies back with an IP address the device could use (DHCP Offer). The device then sends a reply confirming it wants the offered IP Address (DHCP Request), and then lastly, the DHCP server sends a reply acknowledging this has been completed, and the device can start using the IP Address (DHCP ACK).

The lab covers various tasks which are described below with what the tasks are and what we are supposed to do in these.

Task 1: Introducing LAN Topologies

Here the tasks explain various topologies like Ring, Mesh, Bus, Star, etc. what are their advantages and limitations. What causes them to fail and how they work and bypass/destroy them.



Answer the questions below

What does LAN stand for?

Correct Answer

What is the verb given to the job that Routers perform?

Correct Answer

What device is used to centrally connect multiple devices on the local network and transmit data to the correct location?

Correct Answer

What topology is cost-efficient to set up?

Correct Answer

What topology is expensive to set up and maintain?

Correct Answer

Complete the interactive lab attached to this task. What is the flag given at the end?

Correct Answer

Task 2: Primer on Subnetting

Answer the questions below

What is the technical term for dividing a network up into smaller pieces?

Correct Answer

How many bits are in a subnet mask?

Correct Answer

What is the range of a subnet (part) of a subnet mask?

Correct Answer

What address is used to identify the start of a network?

Correct Answer

What address is used to identify devices within a network?

Correct Answer

What is the name used to identify the device responsible for sending data to another network?

Correct Answer

Task 3: ARP

Answer the questions below

What does ARP stand for?

Correct Answer

What category of ARP Packet asks a device whether or not it has a specific IP address?

Correct Answer

What address is used as a physical identifier for a device on a network?

Correct Answer

What address is used as a logical identifier for a device on a network?

Correct Answer

Task 4: DHCP

Answer the questions below

What type of DHCP packet is used by a device to retrieve an IP address?

Correct Answer

What type of DHCP packet does a device send once it has been offered an IP address by the DHCP server?

Correct Answer

Finally, what is the last DHCP packet that is sent to a device from a DHCP server?

Correct Answer

Real-World LAN Example

A college computer laboratory is a practical example of a LAN. Many computers may be connected to one or more switches. The computers can access shared internet connectivity, communicate with internal servers, and use shared printers. The network administrator can apply security controls, monitor traffic, and restrict access to selected resources.

Conclusion

Networking is a basic but important part of cybersecurity. Understanding clients, servers, packets, IP addresses, MAC addresses, switches, and LAN topology makes it easier to study more advanced topics. These fundamentals provide the starting point for learning network security, ethical hacking, monitoring, and troubleshooting.